

4(a)(i)	<i>reject if more than one component given ignore names of regions</i> P – elastic tissue / elastic lamina / elastic fibres / elastin ; bod endothelium / endothelial cells Q – <u>smooth muscle</u> (tissue / cells) ; R – collagen fibres / collagen ;	3
4(a)(ii)	<i>any three from:</i> I thick wall to withstand pressure I names of tunica layers 1 endothelium, for protection / folded to allow expansion / allows smooth blood flow / reduces friction / reduces resistance to flow ; 2 elastic fibres / AW, (stretch to) allow artery to, expand / increase (lumen) diameter ; 3 elastic fibres / AW, recoil to, help (onward) blood flow / help maintain pressure / prevent overstretching / prevent bursting ; 4 smooth muscle contracts to, maintain pressure (for onward blood flow) / reduce blood flow to capillaries / for vasoconstriction I constricts <i>for contracts</i> 5 smooth muscle relaxes, to increase blood flow to capillaries / for vasodilation ; <i>if mp4 and mp5 are not gained, allow one mark for smooth muscle for vasoconstriction and vasodilation or smooth muscle contracts and relaxes to, change diameter of lumen / control blood flow</i> 6 collagen (provides strength to) provide support / maintain structure / withstand pressure or collagen prevents, overstretching / bursting / AW ;	3
4(a)(iii)	<i>any two from: ora for muscular artery</i> 1 (lumen / diameter, of) aorta is, wider / larger / bigger / greater ; 2 thicker, wall / tunica media / region Q (Figure 4.1) ; bod middle layer 3 (wall has) greater proportion of elastic fibres to (smooth) muscle (in tunica media) ; 4 (wall has) more collagen fibres in, tunica externa ; A tunica adventitia bod outer layer 5 lumen is smaller relative to thickness of wall / smaller ratio of lumen : wall thickness / larger ratio of wall thickness : lumen	2

Question	Answer	Marks
4(b)	I any dots in cytoplasm or nucleus I any differences in size <i>neutrophil</i> nucleus divided into at least two lobes ; <i>lymphocyte</i> circular nucleus fills almost all of the cell ; A at least 75% R kidney-shaped nucleus	2
4(c)	any one from: if more than one answer all the answers must be correct <i>bronchus has</i> ciliated epithelium / cilia ; goblet cells / mucous glands ; cartilage ;	1